

13 (v2) — LNG Main Fuel: Per-Fuel CO2 Split (D6 fix #3)

Proves · Per-fuel CO₂: ME and AE burning different fuels must produce two splits that sum to the panel total.

Mechanics this scenario turns on

- ME and AE each use **their own fuel's** CO₂ factor (MGO/MDO 3.93267 · HFO 3.114 · LNG 2.753 · Ammonia 0.35154). The two per-engine cards must sum to the panel total.
- SFOC (g/kWh) depends on **load**: a large 2-stroke burns ~167 at 63 % and over 230 at 5 %. That curve, not the arithmetic, is why running spare engines at low load is expensive.

Panels described below · Battery Contribution — UNCHANGED · Power Demands · Baseline — THE core check: two fuels, two factors, cards must sum · IL1 · Battery Benefit

Anything not described here behaves exactly as in [01-excel-baseline](#) — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 01.

Excel loads, but the plant is LNG-capable: **Dual Fuel Engine id 5, 2×22 000, SG 2 800** · AE 4×4 000 (MGO) · fuel price \$620 (LNG default).

v1 of this file paired LNG with a Liquid-family diesel — the client's fuel-family guard correctly coerced it back to MGO (and reset the price to MGO's default 950). Working as intended; the scenario was fixed to use a Gas-family engine.

Battery Contribution — UNCHANGED

SR **444.7** / PS **204.4**, identical allocation table to test 01: the cascade knows nothing about engines or fuels. Clean module isolation.

Power Demands

Hotel 3 872.2 = SG **2 800** (the smaller Dual-Fuel SG) + AE **1 072**. ME = 11 835.5+2 800 = 14 635.5 (header ≈ 14 636).

Baseline — THE core check: two fuels, two factors, cards must sum

Combos reshuffle on the new SFOC curve: 2.6778 (1/1) / **2.6831 (2/0 — second now!)** / 2.6865 (1/2 ← baseline) / 2.6897 / 2.6914.

- Baseline FOC = 2.6865×5 000 = **13 432.4 t/yr** = ME 12 219.3 + AE 1 213.2
- CO₂ per engine, each with ITS OWN factor: ME 12 219.3 × **2.753** (LNG) = **33 639.6** · AE 1 213.2 × **3.93267** (MGO) = **4 771.1** → total **38 410.7** ✓ (the pre-D6 single-constant bug cannot reproduce this)
- Cost = 13 432.45 × 620 = **\$8 328 116**

IL1

13 389.0 t/yr (savings 43.4 t = \$26 935) · CO2 **38 239.8** = ME 33 639.6 + AE 4 600.2. Note ME CO2 identical in baseline and IL1 — the ME load doesn't change between them.

Battery Benefit

173.4 t/yr = \$107 516 (× the \$620 LNG price) — nearly test 01's tonnage: same cascade.

Takeaway: fuel type never touches the FOC math — it prices CO2 and dollars per engine. Engine type DOES touch FOC (new SFOC curve reshuffles the ranking).

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-10 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)